Five data mistakes that distort payment truth.

The query can run and the journal can balance while the report is still wrong.

A PRACTICAL REVIEW

Precision / grain / replay / time / missing vs zero

Money crosses the boundary as text.

A binary float is already an approximation. Reject that ambiguity before it reaches payment totals.

SAFER INPUT CONTRACT

```
amount_text = "12.34"
amount_minor = to_minor(amount_text, "USD")  # 1234
```

Validate the currency scale. Reject excess fractional digits instead of silently rounding them.

The join multiplied the money.

Two captures joined to two fee rows create four combinations. Summing after the join inflates both sources.

INPUT GRAINS

2 captures x 2 fees
= 4 joined rows

SAFER PATTERN

Aggregate each source
at payment + currency grain

SUM(DISTINCT amount) is not a repair

It also collapses equal-valued legitimate rows.

Delivery can happen more than once.

Webhooks and batches are retried. Design for replay instead of assuming exactly-once arrival.

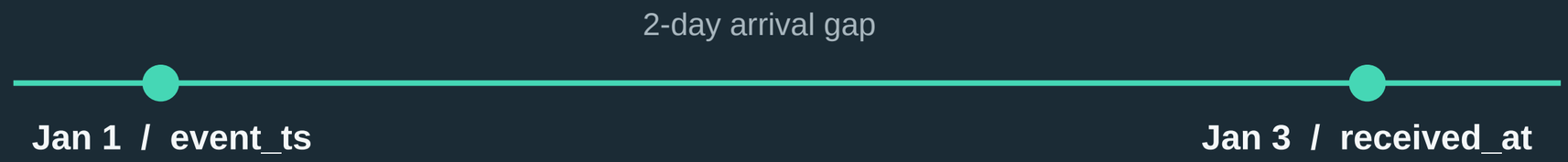
```
evt_42   ->   accepted
evt_42   ->   duplicate; 0 new rows
```

CONTROL

Use a stable idempotency key at the event grain. Test replay and report duplicates explicitly.

Event time is not arrival time.

An event may describe January 1 and only reach the warehouse on January 3.



ASK BOTH

What happened by event time T? What had arrived by cutoff C? Preserve both timestamps.

Missing is not the same as zero.

A settlement-only row has an external amount but no internal expected amount.

SYNTHETIC USD CASE

p9 **expected: unknown** **settlement: 500 cents**

Classify: unexpected external

Avoid COALESCE(expected, 0) unless absence truly means zero.

The five-minute lab uses synthetic data and no paid cloud account.